

Linear regression from scratch

Competition 1 & 2 • Used-car sale price prediction

Data analysis report • September 21, 2026

The selected model uses **unregularized linear regression fitted by gradient descent**. It excludes financing information, handles missing and unfamiliar inputs, and was refitted on all 1,800 public listings. L2 reduced coefficient size but did not produce a meaningful validation improvement.

Development result	Value
Frozen Part I validation RMSE	\$3,683.24
Five additional splits: mean RMSE, $\lambda = 0$	\$3,904.69
Five additional splits: mean RMSE, $\lambda = 0.00001$	\$3,904.67
Final choice	$\lambda = 0$; 49 transformed features; all 1,800 public rows

Public data audit

The supplied CSV has **1,800 rows and 19 columns**. Listing IDs are unique, there are no duplicate rows, and every target is present. Sale prices range from \$1,200.00 to \$75,374.70. The data has 10 numeric candidate predictors and 6 categorical candidate predictors, in addition to the ID, financing field, and target.

Column with missing values	Count	Share
engine_liters	141	7.83%
condition_score	174	9.67%
service_history_score	264	14.67%
fuel_type	41	2.28%
transmission	69	3.83%
loan_approval_value_usd (excluded)	216	12.00%

Odometer readings reach 3,227,800.802 km. Such values deserve investigation, but unusual listings were not automatically removed. Six validation candidates compared alternative mileage treatments. All 1,800 public rows remained eligible for fitting.

Sources and scope

Sources: supplied **T1&2.pdf**, pages 1-4, and **car_price_train.csv**. No external dataset, data generator, hidden features, or hidden labels were used. All scores in this report are public-data development results, not a classroom hidden-test score.

02 Predictors and preprocessing

Prediction occurs after inspection and before negotiation and financing. Without a data dictionary, vehicle descriptors and inspection results are assumed available at that point. This assumption is explicit; the financing-related field is excluded conservatively.

Field or group	Availability and final treatment
listing_id	Identifier only; preserved for joining predictions. Never a predictor.
sale_price_usd	Unknown at inference; training target only.
loan_approval_value_usd	Financing-related; assumed unavailable and excluded.
model_year; vehicle_age_years	Known vehicle descriptors. Both retained.
odometer_km	Available at inspection, but omitted in the selected validation candidate.
annual_mileage_km	Assumed known from recorded vehicle history; retained.
engine_liters; horsepower	Vehicle specifications; both retained.
condition_score; service_history_score	Inspection/history measures; both retained.
prior_owners; accident_count	Recorded history available by inspection; both retained.
brand; body_style; fuel_type; transmission	Known vehicle categories; all retained.
region; seller_type	Known listing categories; both retained.
age × annual mileage	Engineered only from available predictors. A mileage proxy, not a recovered true odometer.

Fitted preprocessing

Numeric values: coerce malformed or infinite values to missing, fill with training medians, and add a missing indicator for each retained original numeric field. An entirely missing training column uses zero as a fallback. Create the selected interaction after filling its inputs.

Categories: strip whitespace, represent missing values as a category, and keep a fixed, sorted training vocabulary. Unseen levels use training category proportions. Their standardized category block is zero, giving the training-average contribution.

Scaling: subtract training means and divide by training standard deviations for numeric features, indicators, and one-hot columns. A constant column uses a scale of one. Training, validation, and test always use the same column order. Every split fits a new preprocessor using its training rows only.

03 Validation design and model mathematics

A seeded random **80/20 holdout** uses 1,440 training rows and 360 validation rows. Seed 42 is used for feature and optimizer development. No listing timestamp or known repeated-vehicle group is supplied; this design estimates performance for similar randomly sampled listings. It is not a time-based generalization test.

Objective and gradients

For standardized feature row x_i , prediction is $\hat{y}_i = x_i^T w + b$. Let $e = Xw + b - y$. The handout requires:

$$J(w, b) = (1/n) \sum_i e_i^2 + \lambda \sum_j w_j^2$$

$$\nabla_w J = (2/n) X^T e + 2\lambda w$$

$$\partial J / \partial b = (2/n) \sum_i e_i$$

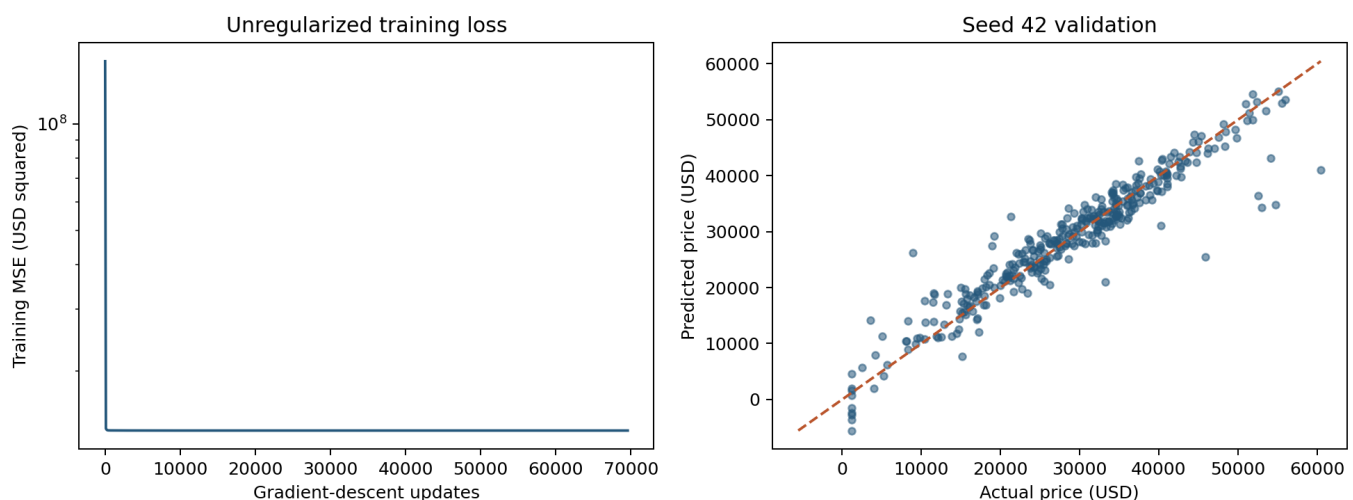
$$w \leftarrow w - \alpha \nabla_w J, b \leftarrow b - \alpha \partial J / \partial b$$

Part I fixes $\lambda = 0$ and therefore minimizes ordinary MSE. Part II adds the specified L2 term. The intercept is never penalized. Scaling matters because a penalty on coefficients depends on feature units.

Implementation and stopping

Initialize all slopes to zero and the intercept to the training mean target. Cache $A^T A / n$ and $A^T y / n$ for $A = [X, 1]$ to calculate the same gradient efficiently. No matrix inverse, least-squares solver, or optimization library is used. A test confirms these updates match direct residual-gradient updates.

Use $\alpha = 0.05$, at most 150,000 updates, and stop when the largest absolute gradient is at most 0.00001. Check for finite parameters, gradients, losses, and predictions. Record the objective every 100 updates and at stopping. Reject a materially increasing objective. The frozen baseline converged after 69,583 updates.



Left: ordinary training MSE decreases and levels off. Right: held-out prices versus raw linear predictions. The dashed line marks perfect predictions. The code does not clip negative predictions.

04 Part I: select and preserve the baseline

Use RMSE as the only prediction metric. Feature candidates share seed 42, initialization, and optimizer settings ($\alpha = 0.05$, maximum 150,000 updates, tolerance 0.001). All six converged. Their small differences should not be treated as strong evidence of a universal feature ranking.

Candidate	Training RMSE	Validation RMSE
Raw odometer	\$3,687.94	\$3,689.42
Training 99th-percentile odometer cap	\$3,682.82	\$3,684.81
Log(1 + odometer)	\$3,673.61	\$3,697.17
Omit odometer	\$3,696.57	\$3,687.09
Replace odometer with age $\times$ annual mileage	\$3,685.23	\$3,683.24
Capped odometer + age squared	\$3,603.81	\$3,686.42

Optimizer investigation on the selected feature set

Rate	Max updates	Tolerance	Updates used	Validation RMSE	Outcome
0.05	150,000	0.001	45,939	\$3,683.24	Converged
0.01	150,000	0.001	150,000	\$3,683.24	Iteration limit
0.1	150,000	0.001	22,967	\$3,683.24	Converged
0.5	150,000	0.001	-	Rejected	Loss increased
0.05	1,000	0.001	1,000	\$3,685.76	Iteration limit
0.05	10,000	0.001	10,000	\$3,683.48	Iteration limit
0.05	50,000	0.001	45,939	\$3,683.24	Converged
0.05	150,000	0.1	22,294	\$3,683.25	Converged
0.05	150,000	1e-05	69,583	\$3,683.24	Converged

Only converged runs were eligible for the frozen baseline. A rate of 0.5 was unstable. Small iteration limits and the slow 0.01 rate did not reach their stopping criteria. Tighter convergence changed RMSE by less than one cent relative to tolerance 0.001; the final setting uses the lowest measured validation RMSE among converged runs.

Preserved baseline

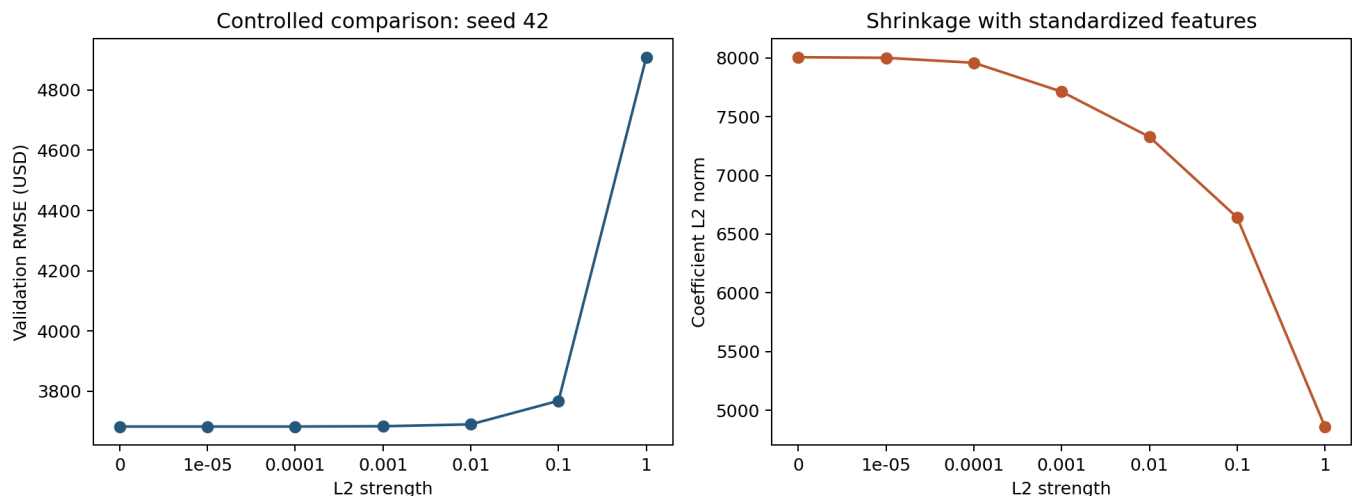
Final Part I: \$3,683.24 validation RMSE; \$3,685.23 training RMSE; coefficient L2 norm 8,005.30. The configuration, parameters, validation predictions, and loss trace were saved in a separate Git commit before the L2 implementation was added.

05 Part II: controlled L2 comparison

The retained predictors **model_year** and **vehicle_age_years** have public-data correlation **-0.997955**. Other observed relationships include engine liters versus horsepower (0.822056) and condition versus service history (0.730071). These motivate testing L2, but correlation by itself does not prove regularization will improve prediction.

Hold the data split, feature set, preprocessing, zero-slope initialization, intercept initialization, learning rate, maximum updates, and tolerance fixed. Change only λ . Every run below converged. The norm excludes the intercept and uses the same standardized feature coordinates.

L2 strength λ	Validation RMSE	Coefficient L2 norm	Updates
0	\$3,683.24	8,005.30	69,583
1e-05	\$3,683.25	8,000.28	69,228
0.0001	\$3,683.34	7,958.08	66,185
0.001	\$3,684.24	7,712.37	45,974
0.01	\$3,690.44	7,327.87	11,337
0.1	\$3,768.66	6,641.89	1,325
1	\$4,908.49	4,862.55	152



What the comparison shows

The best positive value is $\lambda = 0.00001$. On seed 42 its RMSE is very slightly worse than $\lambda = 0$. Larger penalties reduce coefficient norm more strongly but increase validation error. At $\lambda = 1$, RMSE reaches \$4,908.49 while the norm drops to 4,862.55.

Setting $\lambda = 0$ reproduces the saved Part I predictions with a maximum absolute difference of **\$0.00** and identical coefficients. Algebraically, both penalty terms vanish exactly at zero.

06 Repeatability and the final choice

Compare $\lambda = 0$ and the selected positive $\lambda = 0.00001$ using five additional 80/20 split seeds. Each seed refits preprocessing on its own training rows. All settings remain the same within each pair. These seeds were set in code before inspecting their outcomes.

Split seed	RMSE: $\lambda = 0$	RMSE: $\lambda = 0.00001$	Positive minus zero
7	\$3,522.38	\$3,522.37	\$-0.0080
21	\$3,935.02	\$3,934.99	\$-0.0326
84	\$4,119.71	\$4,119.64	\$-0.0717
123	\$4,238.67	\$4,238.69	\$+0.0208
2026	\$3,707.68	\$3,707.68	\$-0.0010

Model	Mean RMSE	Minimum	Maximum
$\lambda = 0$	\$3,904.69	\$3,522.38	\$4,238.67
$\lambda = 1e-05$	\$3,904.67	\$3,522.37	\$4,238.69

Is regularization necessary?

No meaningful predictive need is demonstrated here. The positive penalty wins on four of five additional splits, but its mean advantage is only **\$0.01849**. The unregularized RMSE range spans more than \$700. The smallest positive λ barely shrinks the coefficients; larger penalties trade stronger shrinkage for worse seed-42 predictions.

Keep the **Part I pipeline with $\lambda = 0$** . After reviewing the near-identical scores, differences below \$1 in repeated-split mean RMSE were treated as a practical tie. This post-comparison tie rule is a stated judgment, not a preregistered rule or statistical significance test. Coefficient shrinkage is reported for explanation, not used as the selection metric.

Final refit

Refit only the selected pipeline on all **1,800 public rows**. The final model uses 49 transformed columns and converged in 69,209 updates. Store the fitted preprocessing and model together. The prediction command uses those saved values without adapting to the hidden categories or labels.

Limits of the evidence

Repeated holdouts overlap and share the same public dataset used for development. They do not supply an independent final test or a confidence interval. Hidden-data missingness, new categories, and other distribution changes can worsen performance. Correlated and redundant columns can make individual unregularized coefficients unstable even when predictions remain similar.

07 Verification, reproduction, and submission

Verification	Evidence
Gradient correctness	Finite-difference checks for $\lambda = 0$ and $\lambda > 0$; cached and direct residual updates agree.
Known solutions	Recover a known line; preserve a constant target; recover a toy L2 slope of 2 with intercept 5.
No leakage through preprocessing	Changing validation values does not change fitted medians. Altering ID, target, or loan field does not change features.
Robust transforms	All-missing numeric inputs, unseen categories, an all-missing training field, and reordered columns remain finite and aligned.
Input and output safety	Reject nonfinite training inputs, unstable learning rates, and duplicate IDs. Produce exactly 600 finite predictions with exact headers.
Saved-model command	A separate CLI smoke run loads the final model and preserves all 600 temporary fixture IDs. No real hidden data was used.

All eight automated tests passed. Results are recorded in **results/test_results.txt**; the saved-model command check is recorded in **results/cli_check.json**. The implementation imports only the standard library, NumPy, Pandas, and Matplotlib.

Reproduce with Python 3.12

```
python -m pip install -r requirements.txt
python inspect_data.py
python experiments.py --part 1
python experiments.py --part 2
python main.py --mode train
python -m unittest -v test_pipeline
```

Tested with Python 3.12.14, NumPy 2.3.5, Pandas 2.2.3, and Matplotlib 3.10.8. Run commands from the project directory. Train, test, and prediction paths are configured near the top of main.py.

When the instructor releases the hidden CSV

```
python main.py --mode predict --test hidden_data/car_price_test.csv
```

The program writes **predictions.csv** with exactly **listing_id,sale_price_usd_pred** and one finite prediction per test ID. It checks 600 rows by default. Hidden data and predictions are excluded from Git.

Remaining course requirements

The local repository preserves meaningful commits, including Part I before L2, and is included in the project ZIP. **GitHub publication still requires account access.** The handout refers to the first competition slides for the exact submission format; those slides were not supplied. The real hidden CSV is also pending. Review the core implementation and use the included presentation notes before the September 29 presentation.

AI assistance: OpenAI ChatGPT/Codex generated the code, experiments, tests, and written report under the handout's explicit AI permission. The AI log records decisions and checks. Student understanding and review are still required.